

TOUGGOURT, Algeria

WMO: 605550

Lat: 33.068N

Lon: 6.089E

Elev: 279

StdP: 14.55

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	36.3	38.8	21.5	16.3	62.4	24.6	18.9	58.0	27.7	56.8	23.7	58.3	2.2	180	0.469

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.3	113.2	72.5	110.7	72.0	107.5	71.6	77.8	99.1	76.3	98.2	74.9	97.4	9.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.0	119.4	87.7	69.9	111.0	87.0	67.9	103.7	87.2	41.3	98.7	39.7	97.5	38.5	97.2	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.1	20.6	17.5	DB	31.7	117.7	1.9	1.8	30.4	119.0	29.3	120.1	28.2	121.1	26.8	122.5	(4)
(5)			WB	29.7	81.5	2.0	2.0	28.3	82.9	27.1	84.1	26.0	85.2	24.5	86.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	72.5	52.2	55.5	62.9	70.3	79.5	88.2	94.0	92.7	84.3	74.1	61.5	53.7	(6)	
	DBStd	15.72	4.10	5.31	6.22	6.57	6.71	6.16	4.81	4.52	6.32	6.11	5.50	4.42	(7)	
	HDD50	48	24	9	0	0	0	0	0	0	0	0	1	14	(8)	
	HDD65	1293	397	271	116	21	0	0	0	0	0	4	133	351	(9)	
	CDD50	8264	92	162	399	610	916	1146	1364	1323	1028	748	347	129	(10)	
	CDD65	4033	0	5	49	180	451	696	899	858	578	287	29	1	(11)	
	CDH74	59691	3	70	596	2059	5882	10733	15180	14105	7872	2924	257	10	(12)	
	CDH80	37827	0	13	174	876	3258	7012	10816	9759	4598	1271	49	1	(13)	
(14) Wind	WSAvg	6.9	5.9	6.8	7.9	8.7	8.7	7.7	6.8	6.6	6.5	5.8	5.6	5.4	(14)	
(15) Precipitation	PrecAvg	2.7	0.8	0.2	0.2	0.2	0.1	0.1	0.0	0.1	0.4	0.2	0.2	0.2	(15)	
	PrecMax	6.7	4.0	1.8	1.0	1.5	1.0	0.4	0.5	0.7	1.3	0.9	1.2	1.0	(16)	
	PrecMin	1.0	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	(17)	
	PrecStd	1.6	1.0	0.4	0.3	0.4	0.2	0.1	0.1	0.2	0.3	0.3	0.3	0.3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.0	80.8	91.0	99.6	107.9	114.1	117.2	115.1	109.1	99.9	85.5	74.2	(19)	
		MCWB	55.5	59.1	60.5	64.3	69.6	71.9	73.0	73.4	71.4	67.0	62.1	56.3	(20)	
	2%	DB	68.2	75.3	84.3	93.8	102.5	110.7	114.5	112.3	104.4	95.0	80.3	69.7	(21)	
		MCWB	53.6	55.9	59.6	63.3	68.4	71.1	72.9	72.8	70.8	67.1	60.5	55.1	(22)	
	5%	DB	65.3	71.3	80.3	89.4	98.3	106.7	111.3	109.2	100.8	90.6	76.7	66.9	(23)	
		MCWB	52.4	54.8	57.7	62.1	67.4	70.5	72.5	72.0	70.8	66.3	59.4	54.1	(24)	
	10%	DB	62.7	67.6	76.6	84.6	94.6	102.6	107.8	105.8	97.2	86.8	73.3	64.4	(25)	
		MCWB	51.3	52.8	56.9	61.1	66.7	69.7	71.9	72.5	70.6	65.9	58.1	52.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.0	61.3	64.8	68.7	74.7	77.8	80.2	79.9	77.6	74.1	65.6	59.4	(27)	
		MCDB	69.0	76.2	77.6	87.2	94.2	98.8	101.1	100.3	94.2	85.0	79.0	68.5	(28)	
	2%	WB	55.4	57.9	61.5	65.9	71.7	75.1	77.4	78.0	75.6	71.0	63.5	57.3	(29)	
		MCDB	65.1	72.1	79.3	84.9	94.4	99.1	100.9	99.3	92.7	85.1	76.8	66.2	(30)	
	5%	WB	53.6	55.6	59.7	64.2	69.8	73.3	75.8	76.4	74.1	69.2	61.1	55.6	(31)	
		MCDB	63.0	68.2	76.4	83.2	91.9	97.8	100.8	98.1	92.1	84.3	73.2	64.7	(32)	
	10%	WB	51.9	53.7	58.0	62.5	68.1	71.7	74.3	75.0	72.8	67.4	59.1	53.8	(33)	
		MCDB	61.0	65.6	73.8	81.1	90.2	97.0	100.4	96.9	90.9	82.4	70.7	63.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	22.0	23.3	24.1	24.2	24.3	24.5	25.3	24.4	22.5	22.9	22.4	21.7	(35)	
		MCDBR	25.0	27.6	28.9	29.8	28.6	28.4	29.4	28.4	26.2	27.0	24.9	24.0	(36)	
	5% WB	MCWBR	14.1	13.7	12.7	12.2	11.2	10.0	10.2	9.8	9.5	10.9	11.6	13.3	(37)	
		MCWBR	22.8	24.5	25.4	25.7	25.1	24.9	25.0	24.0	22.4	22.2	23.1	21.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.362	0.393	0.498	0.546	0.604	0.608	0.644	0.610	0.589	0.512	0.419	0.378	(40)	
		taud	2.122	2.021	1.745	1.665	1.576	1.578	1.532	1.597	1.650	1.787	1.972	2.068	(41)	
	Ebn at Noon		263	267	246	241	229	227	218	222	219	224	238	247	(42)	
		Edh at Noon	39	48	69	78	87	86	90	83	75	61	46	40	(43)	
(44) All-Sky Solar Radiation	RadAvg		1112	1418	1760	2105	2282	2425	2397	2164	1815	1459	1148	988	(44)	
	RadStd		49	56	73	82	73	62	37	51	51	65	39	47	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		+0.68	N/A	N/A	N/A	N/A	N/A	-27	N/A	+238	N/A

Nomenclature: See separate page